

Assignment 3

1) Two GAN networks

Network	Input	Output	Goal
Generator	Noise z	Fake image	Fool discriminator
Discriminator	Image	Real/Fake probability	Detect fake images

2) Meaning of $D(x)$

- $0.9 \rightarrow$ very real
- $0.1 \rightarrow$ very fake
- $0.5 \rightarrow$ perfect equilibrium (discriminator confused)
- Ideal equilibrium because the discriminator can't decide whether it's real or not, this means fake distribution almost equals the normal distribution

3) GAN training

1. Train discriminator to detect fake images
 2. Train generator to fool discriminator
- Repeat $\rightarrow$ adversarial game.

4) No single loss curve

Because two models compete $\rightarrow$ losses fluctuate.

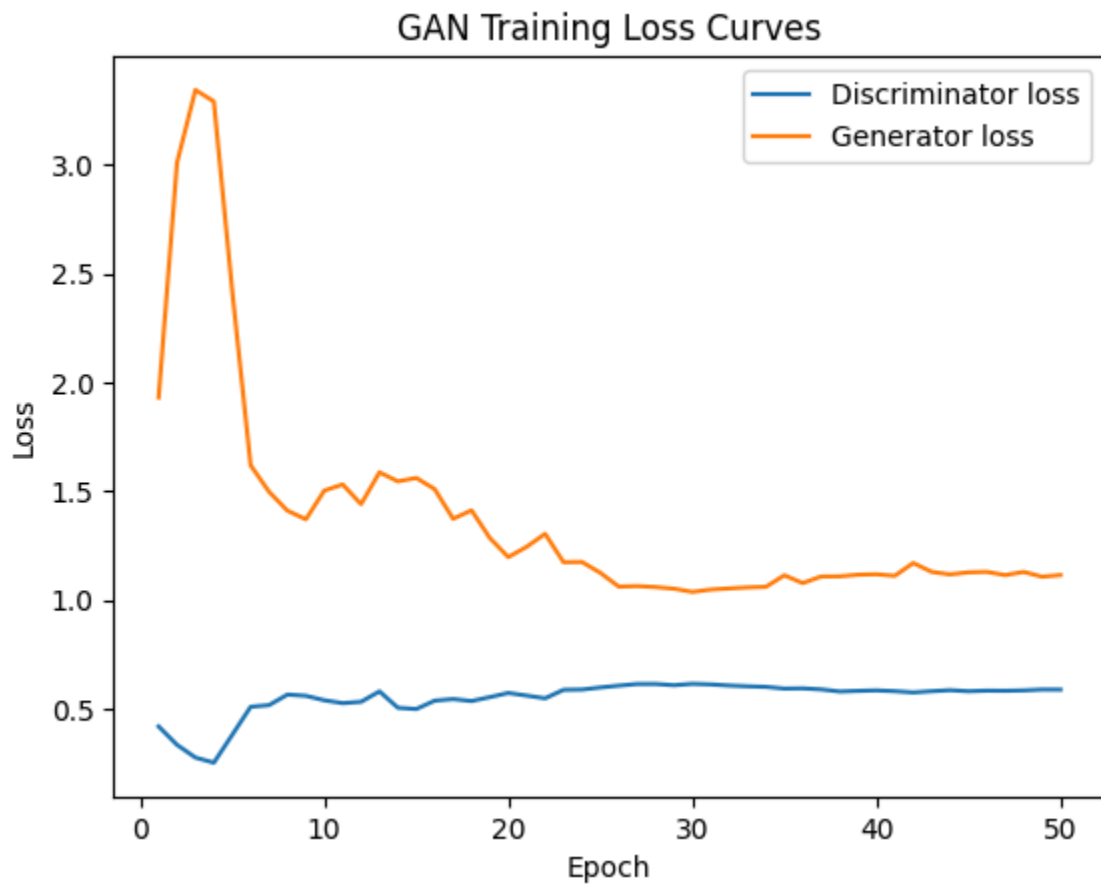
5) Mode collapse

Generator produces repetitive output that fools the discriminator, because generator is rewarded by fooling the discriminator not by covering the full data distribution. If discriminator became too strong, the generator receives near zero gradients and stops learning.

Programming answers

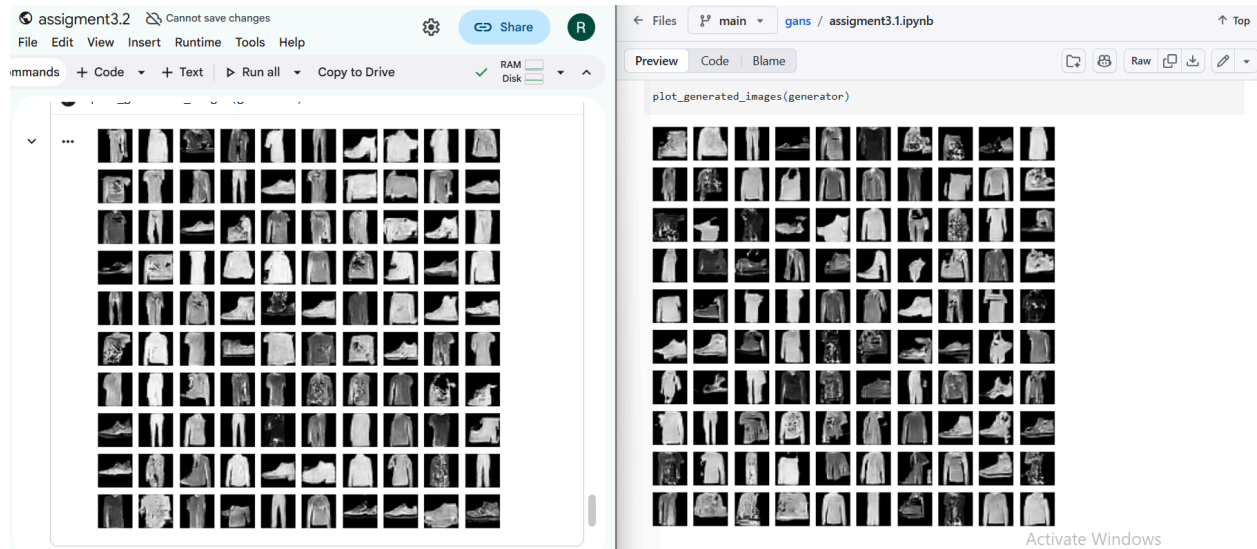
Generator input: random noise vector size 100 → ensures diversity.

Loss behavior: fluctuates → expected because of competition



latent_dim 100 → 10

- Lower quality images.
- Less diversity.



Links for the notebook ;

[assignment3.1](#)

[assignment3.2](#)